Path-level Theorem B: explicit $n = 5$ injection, the refutation of Open Question Q3, and a surprising trace identity

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Abstract

This continues `2026-05-06-theorem-B-paths.tex`, which proved Theorem B at the path level by a pigeonhole existence argument. We now do three things. (i) We exhibit the *explicit* $n = 5$ injection $\iota : \mathrm{Paths}(V_{(4,1)}) \hookrightarrow \mathrm{Paths}(V_{(3,2)}) \sqcup \mathrm{Paths}(V_{(3,2)})$, with all 9 source paths and 18 target slots tabulated; the un-injected complement has multiplicity vector $(1, 7, 1) = \mathrm{atom}_{\mathrm{RTL}}$ on B_4 as predicted. (ii) We refute Open Question Q3 of `2026-05-06-theorem-B-multiset.tex`: the closed W-graph paths of $V_{(3,2,1)}$ that traverse the s_5-coloured edge do *not* have generating function equal to $\mathrm{atom}_{\mathrm{RTL}}$ (or any $q^a(1+q)^b$ multiple thereof). The actual generating function is $G(q) = q^4(1+q)(q^6+14q^5+59q^4+94q^3+59q^2+14q+1)$, palindromic of post-prefix degree 6 rather than 4. (iii) We discover an exact and unexplained trace identity:

$$D(q) \;=\; \mathrm{cross}_{(3,1,1)\to(3,2)}(q),$$

where $D(q)$ is the closed-path trace contribution where step 11 (the s_5 step) acts as $(1+q)D_5$ instead of vM_5, and $\mathrm{cross}_{(3,1,1)\to(3,2)}$ is the S_5-block off-diagonal contribution to $\sigma_1(V_{(3,2,1)})$ from KL-basis indices in the $V_{(3,1,1)}$-summand to those in the $V_{(3,2)}$-summand of the branching $V_{(3,2,1)} \downarrow S_5$. We explore but do not resolve the structural reason for this identity. The path-level data on which all this rests was computed numerically in `~/projects/scratch/2026-05-06-paths/`, including the full Kazhdan–Lusztig μ-graphs of the three cells $C_{(3,2)}, C_{(4,1)} \subset S_5$ and $C_{(3,2,1)} \subset S_6$.

1 Setup

The framework is from `2026-04-24-sigma1-G1.tex` and `2026-05-06-theorem-B-paths.tex`. We recall only the essentials.

For $\lambda \vdash n$, let V_λ be the Kazhdan–Lusztig (KL) cell module on the basis $\{C_w : w \in \mathcal{C}_\lambda\}$ indexed by SYTs of shape λ. Fix the staircase reduced word for $w_0 \in S_n$,

$$\underline{w}_0 \;=\; (1;\ 2,1;\ 3,2,1;\ \ldots;\ n{-}1,\ldots,1), \qquad N := \binom{n}{2}.$$

The Bott–Samelson operator is $\Pi_q := \prod_{j=1}^{N}(T_{i_j}+1)$. In the KL basis, $T_i + 1 = (1+q)D_i + vM_i$ with D_i a diagonal $\{0,1\}$-matrix recording the left-ascent set ($i \notin D_L(w)$) and M_i off-diagonal with μ-coefficient entries. Closed W-graph paths $\pi = (w_0, \ldots, w_N; S; \mathbf{e})$ have bigrading $c = |S^c|/2$,

$d = |S|$ with $2c + d = N$, and

$$\sigma_1(V_\lambda) \;=\; \mathrm{tr}\big(\Pi_q \,|\, V_\lambda\big) \;=\; \sum_{\pi \in \mathrm{Paths}(V_\lambda)} (1+q)^{d(\pi)} q^{c(\pi)}.$$

After dividing σ_1 by its prefix $q^{a_\lambda}(1+q)^{b_\lambda}$ we obtain $A_\lambda(q) \in \mathbb{Z}_{\geq 0}[q]$, palindromic of degree $\deg A_\lambda$. The post-prefix bigrading is $\bar{c} = c - a_\lambda$, $\bar{d} = d - b_\lambda$. The multiplicity vector $M_\lambda = (m_0, \ldots, m_{\deg A_\lambda/2})$ records $m_{\bar{c}} = |\{\pi : \bar{c}(\pi) = \bar{c}\}|$, equivalently the canonical greedy palindromic decomposition of A_λ in the basis

$$B_{2k} \;=\; \big\{q^c(1+q)^{2k-2c} \,:\, c = 0, \ldots, k\big\}.$$

For the present paper the relevant five vectors are

$$M_{(3,2,1)} = (1, 9, 15, 2), \;\; M_{(5,1)} = (1, 3, 9, 14), \;\; M_{(3,1,1)} = (1, 2), \;\; M_{(3,2)} = (1, 5, 3), \;\; M_{(4,1)} = (1, 3, 5). \tag{1}$$

The S_5 staircase has length 10; the S_6 staircase has length 15 and consists of the S_5 word followed by the new block $(s_5, s_4, s_3, s_2, s_1)$ at positions $11, 12, 13, 14, 15$.

Computational basis. All path-level statements in this paper are based on the KL polynomial / W-graph computations in `~/projects/scratch/2026-05-06-paths/` (Sympy, no SageMath). The W-graphs of $C_{(3,2)}, C_{(4,1)} \subset S_5$ and $C_{(3,2,1)} \subset S_6$ were computed by the standard right-multiplication recursion for KL polynomials (Björner–Brenti Theorem 5.1.8). All μ-coefficients in the relevant cells are $\in \{0, 1\}$. Closed paths were enumerated with edge multiplicities, and the resulting $\sigma_1(V_\lambda)$ polynomials were verified to match the closed-form values from `2026-05-06-twin-multiset.tex`.

2 Explicit $n = 5$ injection ι

Path enumeration

The KL W-graph of $C_{(3,2)} \subset S_5$ has 5 vertices (SYTs of shape $(3,2)$), 6 edges, and 9 closed paths along the staircase word for $w_0 \in S_5$. The W-graph of $C_{(4,1)} \subset S_5$ has 4 vertices, 3 edges (a path graph), and also 9 closed paths. In both cases the path enumeration recovers σ_1 exactly:

$$\sigma_1(V_{(3,2)}) = q^2(1+q)^2(q^4 + 9q^3 + 19q^2 + 9q + 1),$$
$$\sigma_1(V_{(4,1)}) = q\,(1+q)^4(q^4 + 7q^3 + 17q^2 + 7q + 1).$$

Distributing paths by post-prefix bigrade $\bar{c} \in \{0, 1, 2\}$:

$\bar{c}$	$\lvert\mathrm{Paths}(V_{(3,2)})_{\bar{c}}\rvert$	$\lvert\mathrm{Paths}(V_{(4,1)})_{\bar{c}}\rvert$	$2\lvert\mathrm{Paths}(V_{(3,2)})_{\bar{c}}\rvert$	complement
0	1	1	2	1
1	5	3	10	7
2	3	5	6	1

The complement column is precisely $\mathrm{atom}_{\mathrm{RTL}}$. Ranges and slack at $\bar{c} = 2$ are tight: $|\mathrm{Paths}(V_{(4,1)})_2| = 5$ exactly fills five of six slots in $\mathrm{Paths}(V_{(3,2)})_2 \sqcup \mathrm{Paths}(V_{(3,2)})_2$.

The injection table

We adopt the canonical lex-order injection: at each bigrade $\bar{c}$, sort the paths in $\mathrm{Paths}(V_{(4,1)})_{\bar{c}}$ and in $\mathrm{Paths}(V_{(3,2)})_{\bar{c}}$ lexicographically by the data $(w_0, S, \text{edge-choices})$, then send the i-th $V_{(4,1)}$-path to the $((i-1) \bmod |\mathrm{Paths}(V_{(3,2)})_{\bar{c}}| + 1)$-th $V_{(3,2)}$-path in copy $\lfloor (i-1)/|\mathrm{Paths}(V_{(3,2)})_{\bar{c}}| \rfloor + 1 \in \{1, 2\}$. The full table is tabulated in `~/projects/scratch/2026-05-06-paths/injection_n5.txt`; here is a summary.

For brevity, we encode each path as $(w_0, S, \mathbf{e})$ with w_0 the start vertex (SYT index in lex order), S the (1-indexed) set of diagonal positions in $\{1, \ldots, 10\}$, and $\mathbf{e}$ the edge-choice tuple (vertex landed on after each non-diagonal step).

Theorem 1 (ι at $n = 5$, explicit)**.** *The following table defines a bigrading-preserving injection* $\iota : \mathrm{Paths}(V_{(4,1)}) \hookrightarrow \mathrm{Paths}(V_{(3,2)}) \sqcup \mathrm{Paths}(V_{(3,2)})$.

- ***Bigrade*** $\bar{c} = 0$**.** *The unique* $V_{(4,1)}$*-path* $(w_0 = 0, S = \{1, 2, 3, 5, 6, 8, 9, 10\}, \mathbf{e} = (1, 0))$ *maps to copy 1 image of the unique* $V_{(3,2)}$*-path* $(0, \{1, 3, 5, 6, 9, 10\}, (1, 0, 1, 0))$*. Complement: copy 2 of* $(0, \{1, 3, 5, 6, 9, 10\}, (1, 0, 1, 0))$*. (1 path.)*

- ***Bigrade*** $\bar{c} = 1$**.** *The 3* $V_{(4,1)}$*-paths inject into copy 1 of the first 3* $V_{(3,2)}$*-paths (lex-sorted). Complement: copy 1 of* $V_{(3,2)}$*-paths 3, 4 (i.e.* $(1, \{3, 5, 8, 10\}, \ldots)$ *and* $(2, \{3, 6, 7, 10\}, \ldots)$*) and all 5* $V_{(3,2)}$*-paths in copy 2. (7 paths.)*

- ***Bigrade*** $\bar{c} = 2$**.** *The 5* $V_{(4,1)}$*-paths inject as follows: the first 3 (lex order) into copy 1 of* $V_{(3,2)}$*-paths 0, 1, 2; the remaining 2 into copy 2 of* $V_{(3,2)}$*-paths 0, 1. Complement: copy 2 of* $V_{(3,2)}$*-path 2, namely* $(2, \{3, 10\}, (4, 2, 4, 2, 4, 3, 4, 2))$*. (1 path.)*

Proof. By direct enumeration. Path counts and bigrades match the multiplicity vectors $(1, 5, 3)$ and $(1, 3, 5)$. The injection is a function: 9 distinct sources, 9 distinct targets in 18 slots. Bigrading is preserved by construction (sort within each $\bar{c}$-bin and inject within bin). □

Remark 2 (Where the choice lives)**.** At $\bar{c} = 1$, only 3 of 10 slots receive an image; the choice of which 3 is unconstrained (we picked the first 3 of copy 1 by lex order, but any 3-element subset of the 10 slots would work). At $\bar{c} = 2$, only 1 of 6 slots is unfilled; the un-filled slot, the lone atom-element here, is specified up to the lex convention. At $\bar{c} = 0$, also 1 of 2 slots unfilled. So the injection is canonical only up to the choice of total order on each bigrade-fibre. Section 5 returns to whether a more natural choice exists.

Lifting to $n = 6$ via convolution

By the proof of Theorem 1 of `2026-05-06-theorem-B-paths.tex`, the convolution $f := \mathrm{id}_{\mathrm{Paths}(V_{(3,1,1)})} \times \iota$ defines a bigrading-preserving injection

$$f : \mathrm{Paths}(V_{(3,1,1)}) \times \mathrm{Paths}(V_{(4,1)}) \ \hookrightarrow\ \mathrm{Paths}(V_{(3,1,1)}) \times \mathrm{Paths}(V_{(3,2)}) \sqcup \mathrm{Paths}(V_{(3,1,1)}) \times \mathrm{Paths}(V_{(3,2)}),$$

whose complement has multiplicity vector $M_{(3,1,1)} * \mathrm{atom}_{\mathrm{RTL}} = (1, 2) * (1, 7, 1) = (1, 9, 15, 2) = M_{(3,2,1)}$.

So Theorem 1 is the genuine path-level content. The $n = 6$ statement is mechanical from it.

3 Refutation of Open Question Q3

`2026-05-06-theorem-B-multiset.tex` closes with the question:

"Is $\mathrm{atom}_{\mathrm{RTL}}$ *the generating function over the closed W-graph paths of $V_{(3,2,1)}$ that traverse the s_5-coloured edge?"*

We answer: **No, not as stated, and not for any simple $q^a(1+q)^b$ rescaling.**

The s_5-step split

Position 11 in the staircase word for $w_0 \in S_6$ is the s_5 letter. Decompose $T_5 + 1 = (1+q)D_5 + vM_5$ in the KL basis, and split each closed path of $V_{(3,2,1)}$ by whether step 11 uses $(1+q)D_5$ ("diagonal", $11 \in S$) or vM_5 ("traverses the s_5 edge", $11 \notin S$).

Theorem 3 (Q3 refuted)**.** *Define*

$$D(q) := \mathrm{tr}\big(\Pi_q^{S_5} \cdot (1+q)D_5 \cdot R_5' \,|\, V_{(3,2,1)}\big) \quad (\textit{step 11 diagonal}),$$
$$G(q) := \mathrm{tr}\big(\Pi_q^{S_5} \cdot vM_5 \cdot R_5' \,|\, V_{(3,2,1)}\big) \quad (\textit{step 11 traverses the } s_5 \textit{ edge}),$$

where $R_5' := (T_4+1)(T_3+1)(T_2+1)(T_1+1)$. Then $D(q) + G(q) = \sigma_1(V_{(3,2,1)})$ and

$$D(q) = q^5\,(1+q)^3\,(q^2+5q+1),$$
$$G(q) = q^4\,(1+q)\,(q^6+14q^5+59q^4+94q^3+59q^2+14q+1).$$

In particular, for every $a, b \in \mathbb{Z}_{\geq 0}$, neither $D(q)/[q^a(1+q)^b]$ nor $G(q)/[q^a(1+q)^b]$ equals $\mathrm{atom}_{\mathrm{RTL}}(q) = q^4 + 11q^3 + 21q^2 + 11q + 1$*.*

Proof. Direct computation in the KL basis, using the explicit W-graph and the D_5, M_5 matrices. The trace was computed by Lagrange interpolation on integer evaluations of the matrix product (script `wgraph_fast.py`, output `main_output.txt`; verified to recover $\sigma_1(V_{(3,2,1)}) = q^4(1+q)(q^6 + 15q^5 + 66q^4 + 106q^3 + 66q^2 + 15q + 1)$).

For the non-equality with $\mathrm{atom}_{\mathrm{RTL}}$: the polynomial $G(q)/[q^4(1+q)] = q^6 + 14q^5 + 59q^4 + 94q^3 + 59q^2 + 14q + 1$ is palindromic of degree 6, with palindromic decomposition $(1, 8, 12, 2)$ on B_6. This is not equal to $\mathrm{atom}_{\mathrm{RTL}}$, which is degree 4 with decomposition $(1, 7, 1)$. No further factor of $(1+q)$ divides $G(q)/[q^4(1+q)]$ exactly, and dividing by any $q^a(1+q)^b$ either leaves a polynomial of wrong degree or fails. ☐

So the picture in Q3 was wrong: the s_5-edge-traversing paths are *more numerous* than the atom predicts, and they have the wrong palindromic shape.

Path counts

Translating to the post-prefix bigrade on $A_{(3,2,1)} = \sigma_1(V_{(3,2,1)})/[q^4(1+q)]$:

$\bar{c}$	0	1	2	3	total
$M_{(3,2,1)}$ (all paths)	1	9	15	2	27
step 11 diagonal D	0	1	3	0	4
step 11 traverses G	1	8	12	2	23

Of the 27 closed paths in $V_{(3,2,1)}$, only 4 have step 11 diagonal — a striking minority. The atom $\mathrm{atom}_{\mathrm{RTL}}$, by contrast, has 9 elements. So the cardinalities don't even match: $|\mathrm{atom}_{\mathrm{RTL}}| = 9$, $|D|_{\mathrm{paths}}| = 4$, $|G|_{\mathrm{paths}}| = 23$. The atom is not on either side of the s_5 split.

4 The surprising trace identity

Although Q3 fails, an unexpected and exact identity emerged from the data. We state it, prove it numerically, and discuss what it might mean.

Setup: the S_5-block decomposition

The branching $V_{(3,2,1)} \downarrow S_5 \;=\; V_{(3,2)} \oplus V_{(3,1,1)} \oplus V_{(2,2,1)}$ partitions the 16 SYTs of shape $(3,2,1)$ by the location of the box containing 6:

μ	vertex indices	condition
$(3,2)$	$\{0,2,4,8,10\}$	6 in row 2
$(3,1,1)$	$\{1,3,6,9,12,14\}$	6 in row 1
$(2,2,1)$	$\{5,7,11,13,15\}$	6 in row 0

Crucially, the KL basis of $V_{(3,2,1)}$ at S_6 does *not* respect this block decomposition under the S_5-action: the matrices M_1, M_2, M_3, M_4 (the off-diagonals of $T_i + 1$ for $i \leq 4$) all have nonzero entries between distinct μ-blocks. Consequently $\Pi_q^{S_5}$ has off-block matrix entries.

The cross-trace decomposition

For each ordered pair (μ_i, μ_j) of partitions in $\{(3,2),(3,1,1),(2,2,1)\}$, define

$$\mathrm{cross}_{\mu_i \to \mu_j}(q) \;:=\; \sum_{\substack{i \in V_{\mu_i} \\ j \in V_{\mu_j}}} (\Pi_q^{S_5})_{ij} \cdot (R_5)_{ji}.$$

By the trace formula $\mathrm{tr}(AB) = \sum_{i,j} A_{ij} B_{ji}$ applied to $A = \Pi_q^{S_5}$ and $B = R_5$, we have

$$\sigma_1(V_{(3,2,1)}) \;=\; \sum_{(\mu_i,\mu_j)} \mathrm{cross}_{\mu_i \to \mu_j}(q).$$

Direct computation in the KL basis gives:

- Diagonal blocks ($\mu_i = \mu_j$):

$$\begin{aligned}
\mathrm{cross}_{(3,2)\to(3,2)} &= q^4 + 14q^5 + 64q^6 + 130q^7 + 130q^8 + 64q^9 + 14q^{10} + q^{11}, \\
\mathrm{cross}_{(3,1,1)\to(3,1,1)} &= q^5 + 7q^6 + 16q^7 + 16q^8 + 7q^9 + q^{10}, \\
\mathrm{cross}_{(2,2,1)\to(2,2,1)} &= q^6 + 3q^7 + 3q^8 + q^9.
\end{aligned}$$

- Off-diagonal blocks ($\mu_i \neq \mu_j$): only *two* of the six are nonzero,

$$\begin{aligned}
\mathrm{cross}_{(3,1,1)\to(3,2)} &= q^5 + 8q^6 + 19q^7 + 19q^8 + 8q^9 + q^{10}, \\
\mathrm{cross}_{(2,2,1)\to(3,2)} &= q^6 + 4q^7 + 4q^8 + q^9.
\end{aligned}$$

The other four off-diagonal blocks ($\mu_i \neq \mu_j$ with $\mu_j \neq (3,2)$) are exactly 0 as polynomials in q. The cross-flux flows entirely *into* the $V_{(3,2)}$-summand from the other two, never out.

The identity

Theorem 4 (Surprising identity)**.**

$$D(q) \;=\; \mathrm{cross}_{(3,1,1)\to(3,2)}(q) \;=\; q^5 + 8q^6 + 19q^7 + 19q^8 + 8q^9 + q^{10}.$$

Computational verification. Both sides equal $q^5 + 8q^6 + 19q^7 + 19q^8 + 8q^9 + q^{10}$ identically. This is verified by independent direct computation in `wgraph_fast.py` (Lagrange-interpolated trace at ~ 18 integer values of q). □

Remark 5 (Why this is surprising)**.** The two definitions are completely different.

- $D(q)$ is the trace contribution where the s_5 step at position 11 acts as the *diagonal* part $(1+q)D_5$ of $T_5 + 1$, regardless of where in the KL basis the path lives. It is a constraint on a single step out of 15.
- $\mathrm{cross}_{(3,1,1)\to(3,2)}(q)$ is the trace contribution from those KL-basis indices (i, j) with $i \in V_{(3,1,1)}$ and $j \in V_{(3,2)}$. It is a constraint on *which μ-summands* the start and end of the $\Pi_q^{S_5}$-arc and the R_5-arc lie in.

There is no a priori reason the two should agree.

A consequence: $G(q)$ in terms of the μ-block decomposition

Combining Theorem 4 with $\sigma_1 = D + G = \sum_{(\mu_i,\mu_j)} \mathrm{cross}_{\mu_i\to\mu_j}$, and the fact that only $(3,1,1) \to (3,2)$ and $(2,2,1) \to (3,2)$ off-diagonal blocks are nonzero,

$$G(q) \;=\; \mathrm{cross}_{(3,2)\to(3,2)} + \mathrm{cross}_{(3,1,1)\to(3,1,1)} + \mathrm{cross}_{(2,2,1)\to(2,2,1)} + \mathrm{cross}_{(2,2,1)\to(3,2)}.$$

So while G is not $\mathrm{atom}_{\mathrm{RTL}}$, it has an exact decomposition into the diagonal blocks plus a single off-diagonal block (the one from the smallest summand $V_{(2,2,1)}$ into the largest $V_{(3,2)}$). This is also a clean structural statement, just not the one Q3 anticipated.

5 What is open

Canonical ι

The injection ι at $n = 5$ (Theorem 1) is constructive but not canonical, in that the lex-order convention is arbitrary. A canonical ι would, for instance, exploit the S_5-Mullineux involution or the branching $V_{(3,2)} \downarrow S_4, V_{(4,1)} \downarrow S_4 \supset V_{(3,1)}$. Neither has been pinned down.

The slack at $\bar{c} = 2$ as a single distinguished path

Theorem 1 singles out the $V_{(3,2)}$-path

$$(w_0 = 2,\; S = \{3, 10\},\; \mathbf{e} = (4, 2, 4, 2, 4, 3, 4, 2))$$

as the unique unfilled slot at $\bar{c} = 2$ (in copy 2). Identifying this path canonically — e.g., the unique path with a specific structural feature in the W-graph of $V_{(3,2)}$ — would substantially constrain the canonical injection. This single path encodes the entire $\bar{c} = 2$ contribution of $\mathrm{atom}_{\mathrm{RTL}}$.

The structural meaning of $D(q) = \mathrm{cross}_{(3,1,1)\to(3,2)}$

Theorem 4 is a numerical identity. The structural reason — if there is one — is unclear. Some observations:

- The s_5-ascent vertices in $V_{(3,2,1)}$ partition by branching as $\{1, 3, 9\}$ in $V_{(3,1,1)}$ (the s_5-ascents inside $V_{(3,1,1)}$) and $\{5, 7, 11, 13, 15\}$ in $V_{(2,2,1)}$ (the entire $V_{(2,2,1)}$-summand). The s_5-descent vertices are $\{0, 2, 4, 8, 10\}$ in $V_{(3,2)}$ (entire summand) and $\{6, 12, 14\}$ in $V_{(3,1,1)}$.
- Equivalently: $V_{(3,2)}$ vertices have 6 in row 2, hence 5 above 6, so $5 \in D_L$ always (descent). $V_{(2,2,1)}$ vertices have 6 in row 0, hence 5 below 6 or beside it, so $5 \notin D_L$ always (ascent). $V_{(3,1,1)}$ vertices have 6 in row 1; 5 can be above (descent) or below (ascent), hence the split.
- So the s_5-edge structure is forced by the branching: $V_{(3,2)}$ is the "descent summand", $V_{(2,2,1)}$ is the "ascent summand", and $V_{(3,1,1)}$ contains both.
- $D(q)$ counts paths with step 11 fixed at an ascent vertex (anywhere in $\{1, 3, 9, 5, 7, 11, 13, 15\}$, which spans $V_{(3,1,1)}$-asc $\cup V_{(2,2,1)}$). $\mathrm{cross}_{(3,1,1)\to(3,2)}$ counts (somewhat) the off-block flow from $V_{(3,1,1)}$ to $V_{(3,2)}$ via the $\Pi_q^{S_5}$-arc.
- These two index different things, and their numerical equality is unexplained.

A clean structural proof, if one exists, would presumably use a symmetry of the W-graph or an interpolation between the two trace decompositions. We leave this as an open question.

What this rules out for Q3-style conjectures

A naive "s_5-edge" Q3 fails. So does a refined "either edge of $T_4 + 1$ as well" (we did not test it formally, but the available cross-μ structure — only flowing into $(3, 2)$ — suggests no simple s_i-step split will yield $\mathrm{atom}_{\mathrm{RTL}}$). The structural meaning of $\mathrm{atom}_{\mathrm{RTL}}$ likely lives at the post-prefix level, not at the level of the S_6 staircase paths directly.

6 Computational summary

All numerical claims in this paper rest on the scripts in `~/projects/scratch/2026-05-06-paths/`, which implement KL polynomial computation for S_n ($n \leq 6$), construct the W-graphs of cells C_λ, enumerate closed paths along the staircase reduced word, and compute the relevant trace decompositions. Cross-checks performed:

- $\sigma_1(V_{(3,2)}), \sigma_1(V_{(4,1)}), \sigma_1(V_{(3,2,1)})$ from path enumeration match the closed-form formulas verified in `2026-05-06-twin-multiset.tex` and `2026-05-06-theorem-B-multiset.tex`.
- Multiplicity vectors $M_{(3,2)} = (1, 5, 3)$, $M_{(4,1)} = (1, 3, 5)$, $M_{(3,2,1)} = (1, 9, 15, 2)$ recovered from path bigrading.
- The W-graph satisfies the Hecke braid and commutation relations for $i \leq 4$ (verified).
- $D(q) + G(q) = \sigma_1(V_{(3,2,1)})$ for the s_5-step split (verified).
- $\sigma_1(V_{(3,2,1)}) = \sum_{(\mu_i,\mu_j)} \mathrm{cross}_{\mu_i\to\mu_j}$ for the KL-block decomposition (verified).
- $D(q) = \mathrm{cross}_{(3,1,1)\to(3,2)}(q)$ as polynomials in q (verified by polynomial subtraction).

Honest assessment

This paper does three things and falls short of two larger goals.

What was achieved.

1. The explicit $n = 5$ injection ι is now tabulated path-by-path (Theorem 1), upgrading `2026-05-06-theorem-B-p` pigeonhole existence to a constructive statement.
2. Open Q3 of `2026-05-06-theorem-B-multiset.tex` is refuted: the s_5-edge-traversing closed paths of $V_{(3,2,1)}$ do not have generating function $\mathrm{atom}_{\mathrm{RTL}}$ or any $q^a(1+q)^b$-multiple thereof. The structural meaning of $\mathrm{atom}_{\mathrm{RTL}}$, if any, lies elsewhere.
3. An exact and unexpected trace identity $D(q) = \mathrm{cross}_{(3,1,1)\to(3,2)}(q)$ was discovered. Its structural significance is open.

What was *not* achieved.

1. A canonical ι, in the strong sense (a W-graph-natural distinguished injection) is still open. Theorem 1 is concrete but uses a lex-order arbitrary choice.
2. The structural meaning of $\mathrm{atom}_{\mathrm{RTL}}$ at the path level remains unidentified. Q3 was the most natural conjecture; it is wrong. We have no alternative.
3. The structural reason for Theorem 4 is unexplained.

The primary value of this writeup is therefore (i) negative (Q3 closed) and (ii) catalytic (Theorem 4 suggests that a deeper $S_n \downarrow S_{n-1}$-branching identity is at play in the σ_1-G1 framework). The next session should investigate whether $\mathrm{cross}_{(3,1,1)\to(3,2)}$ generalizes to other (λ, n), and whether the identity $D = \mathrm{cross}_{\mu\to\nu}$ has a cohomological or recursive proof.